Esther Rituerto-González, Dr.

Data Scientist & Researcher

I am an enthusiastic and dynamic individual, committed to driving positive change within society. With a strong background in Artificial Intelligence and a genuine passion for harnessing its potential for social good, I am motivated to channel my knowledge and abilities towards finding innovative solutions that make a lasting impact.

info & contact

esther.rituerto.g@gmail.com

Von-Paris-Straße,
Augsburg, Germany

0034-639-259-597

25/September/1995

portfolio

github.com/erituert/

linkedin.com/in/esther-rituerto-gonzalez/

Personal site: erituert.github.io

Google Scholar Profile

@erituert

languages

English: Proficent level (C2)

German: Intermediate level (B1)

Spanish: Native language

interests

Artificial Intelligence

Social Good

Affective Computing

Healthcare and Wellbeing

education

April 2019 - Jan 2023

PhD in Multimedia and Communications

University Carlos III of Madrid (UC3M), Spain

Presented the 'Cum Laude' doctoral thesis "Multi-modal Affective Computing in Wearable Devices with Applications in the Detection of Gender-based Violence" where I applied machine and deep learning algorithms for speaker recognition and emotion detection

2017 - 2018

Master of Research in Multimedia and Communications

University Carlos III of Madrid (UC3M), Spain

Specialization in multimedia data and signal processing, speech technologies, computer vision, machine and deep learning

2013 - 2017

Engineering Degree in Audiovisual Systems

University Carlos III of Madrid (UC3M), Spain

Development of networks, services and multimedia applications in communications networks; acoustic design of enclosures

work experience

April 2023 - Present

Visiting Postdoctoral Scholar

University of Augsburg, Germany

In the Chair of Embedded Intelligence for Health Care and Wellbeing, under the supervision of Prof. Björn Schuller, researching on robust automatic acoustic event detection for real-life scenarios

Feb 2023 - Present

Postdoctoral Researcher

University Carlos III of Madrid (UC3M), Spain

Researching on the detection of PTSD via speech using adversarial deep learning strategies in speaker-independent frameworks

April 2019 - Jan 2023

Predoctoral Researcher

University Carlos III of Madrid (UC3M), Spain

Working in the EMPATIA-CM Project for the development of BINDI, a wearable device powered by Artificial Intelligence to detect, prevent and combat Gender-based Violence.

2021 (6 months)

Visiting Predoctoral Scholar

University of Augsburg, Germany

In the Chair of EIHW, under the supervision of Prof. Björn Schuller, researching on the speech biomarkers of emotions

2017 (6 months)

Internship Student

IMDEA Networks Research Institute in Madrid, Spain

Web portals maintenance, upgrades and improvements, providing support to Engineering, Operations and Communications Departments

research & academia

- Authored and co-authored **13 Journal and Conference publications** such as:
  - BINDI: Affective Internet of Things to Combat Gender-Based Violence. Jose A. Miranda, Esther Rituerto-González, Clara Luis-Mingueza, Manuel F. Canabal, Alberto Ramírez Bárcenas, Jose M. Lanza-Gutiérrez, Carmen Peláez-Moreno, Celia López-Ongil. In IEEE Internet of Things, 2022. [**paper**] [**github_code**]
  - Prediction of the Gender-based Violence Victim Condition using Speech: What do Machine Learning Models rely on?. Emma Reyner-Fuentes, Esther Rituerto-González, Isabel Trancoso, Carmen Peláez-Moreno. In INTERSPEECH 2023 Proceedings. [accepted]
  - Bridging the Semantic Gap with Affective Acoustic Scene Analysis: an Information Retrieval-based Approach. Esther Rituerto-González , Clara Luis-Mingueza, Carmen Peláez Moreno. In IBERSPEECH 2022 Proceedings. [**paper**] [**github_code**]
  - Fatigue Prediction in Outdoor Running Conditions using Audio Data. Andreas Triantafyllopoulos, Sandra Ottl, Alexander Gebhard, Esther Rituerto-González, Mirko Jaumann, Steffen Hüttner, Valerie Dieter, Patrick Schneeweiß, Inga Krauß, Maurice Gerczuk, Shahin Amiriparian, and Björn W. Schuller. In Proceedings of International Conference on Engineering in Medicine and Biology Conference (EMBC), 2022. [**paper**]

- Attendance to **16 Conferences** and Workshops and **3 Machine and Deep Learning Summer Schools**

- Reviewer in 2 **ACM Multimedia and ICML Competitions**.

- More than **150 teaching hours at UC3M** lecturing in Machine Learning, Digital Image Processing and Acoustics related subjects

general skills

Deep Learning | Machine Learning | Data Visualization | Statistical Analysis | Signal Processing | Data Analysis | Statistical Computing | Data Processing

it & technical skills

- Python (PyTorch, Keras, TensorFlow, scikit-learn, numpy, pandas, matplotlib)
- MATLAB
- GitHub, GitLab
- LaTeX
- Java
- C
- HTML, CSS, Javascript
- Ruby on Rails

awards & scholarships

- 2021: Short-stay Mobility Grant for members of the FPU program – MICIU, Spain
- 2021: Short-term Predoctoral Grant – German Academic Exchange Service (DAAD)
- 2019: University Teacher In Training, Predoctoral Research Scholarship (FPU Program 19/00448) Acceptance Rate=37% – Ministry of Science, Innovation And Universities (MICIU), Spain
- 2019: Predoctoral Research Personnel in Training – Community of Madrid YEI Predoctoral Program (PEJD-2019-PRE/TIC-16295), Spain
- 2019: IV Edition Pilar Azcárate Award in Gender-based Violence Research for Master's Thesis – UC3M
- 2017: Complete Financial Aid to Master Studies – UC3M
- 2013: Excellence Award for Best Grades in High School – UC3M